Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

NetApp Solutions

NetApp
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Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

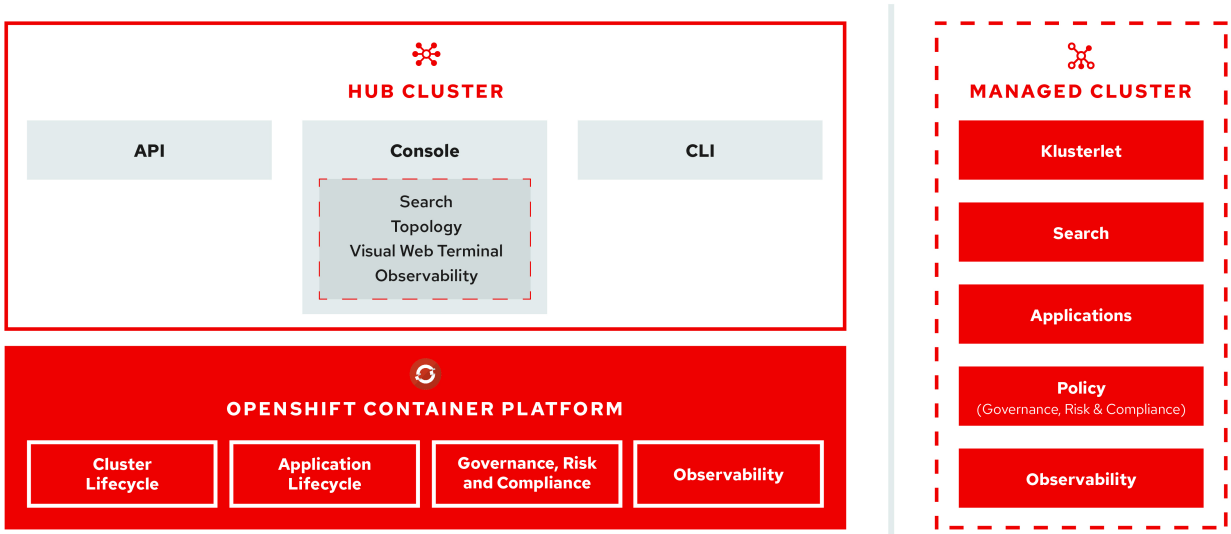
Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

As a containerized application transitions from development to production, many organizations require multiple Red Hat OpenShift clusters to support the testing and deployment of that application. In conjunction with this, organizations usually host multiple applications or workloads on OpenShift clusters. Therefore, each organization ends up managing a set of clusters, and OpenShift administrators must thus face the added challenge of managing and maintaining multiple clusters across a range of environments that span multiple on-premises data centers and public clouds. To address these challenges, Red Hat introduced Advanced Cluster Management for Kubernetes.

Red Hat Advanced Cluster Management for Kubernetes allows the users to:

1. Create, import, and manage multiple clusters across data centers and public clouds.
2. Deploy and manage applications or workloads on multiple clusters from a single console.
3. Monitor and analyse health and status of different cluster resources.
4. Monitor and enforce security compliance across multiple clusters.

Red Hat Advanced Cluster Management for Kubernetes is installed as an add-on to a Red Hat OpenShift cluster, and it uses this cluster as a central controller for all its operations. This cluster is known as hub cluster, and it exposes a management plane for the users to connect to Advanced Cluster Management. All the other OpenShift clusters that are either imported or created via the Advanced Cluster Management console are managed by the hub cluster and are called managed clusters. It installs an agent called Klusterlet on the managed clusters to connect them to the hub cluster and serve the requests for different activities related to cluster lifecycle management, application lifecycle management, observability, and security compliance.



For more information, see the documentation here.

Next: Deployment Prerequisites.

Deployment

Deploy Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

Prerequisites

1. A Red Hat OpenShift cluster (greater than version 4.5) for hub cluster
2. Red Hat OpenShift clusters (greater than version 4.4.3) for managed clusters
3. Cluster-admin access to Red Hat OpenShift cluster
4. A Red Hat subscription for Advanced Cluster Management for Kubernetes

Advanced Cluster Management is an add-on on for the OpenShift cluster, so there are certain requirements and restrictions on the hardware resources based on the features used across the hub and managed clusters. You need to take these issues into account when sizing the clusters. See the documentation here for more details.

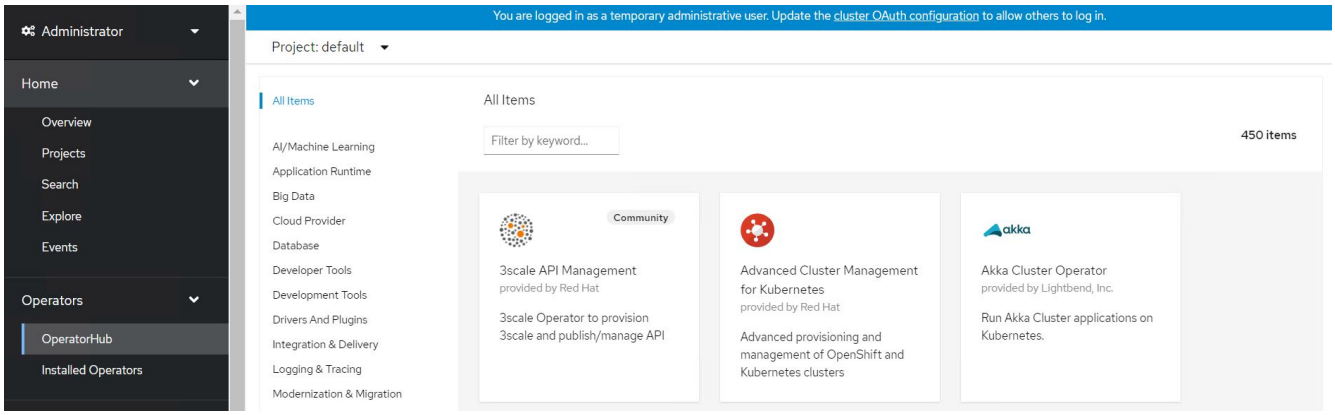
Optionally, if the hub cluster has dedicated nodes for hosting infrastructure components and you would like to install Advanced Cluster Management resources only on those nodes, you need to add tolerations and selectors to those nodes accordingly. For more details, see the documentation here.

Next: Installation.

Deploy Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

To install Advanced Cluster Management for Kubernetes on an OpenShift cluster, complete the following steps:

1. Choose an OpenShift cluster as the hub cluster and log into it with cluster-admin privileges.
2. Navigate to `Operators` → `Operators Hub` and search for `Advanced Cluster Management for Kubernetes`.



3. Select the `Advanced Cluster Management for Kubernetes` and click `Install`.

4. On the Install Operator screen, provide the necessary details (NetApp recommends retaining the default parameters) and click `Install`.

OperatorHub > Operator Installation

Install Operator

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel *

- release-2.0
- release-2.1
- release-2.2

Installation mode *

- All namespaces on the cluster (default)
  This mode is not supported by this Operator
- A specific namespace on the cluster
  Operator will be available in a single Namespace only.

Installed Namespace *

- Operator recommended Namespace: PR open-cluster-management

Namespace creation
Namespace **open-cluster-management** does not exist and will be created.

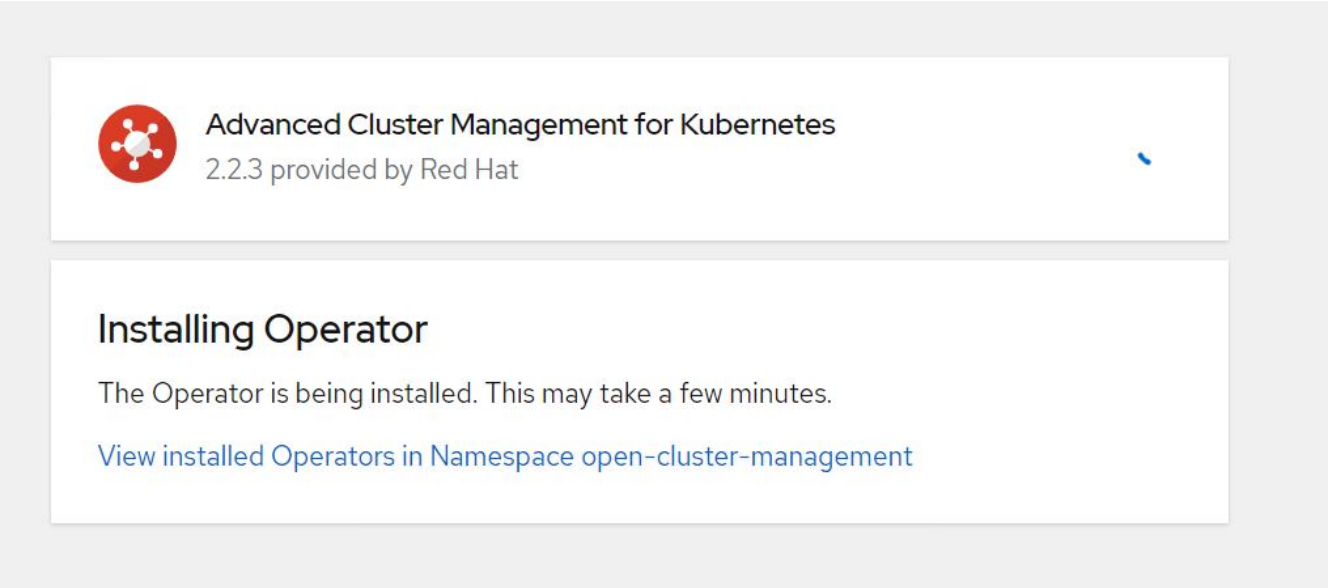
- Select a Namespace

Approval strategy *

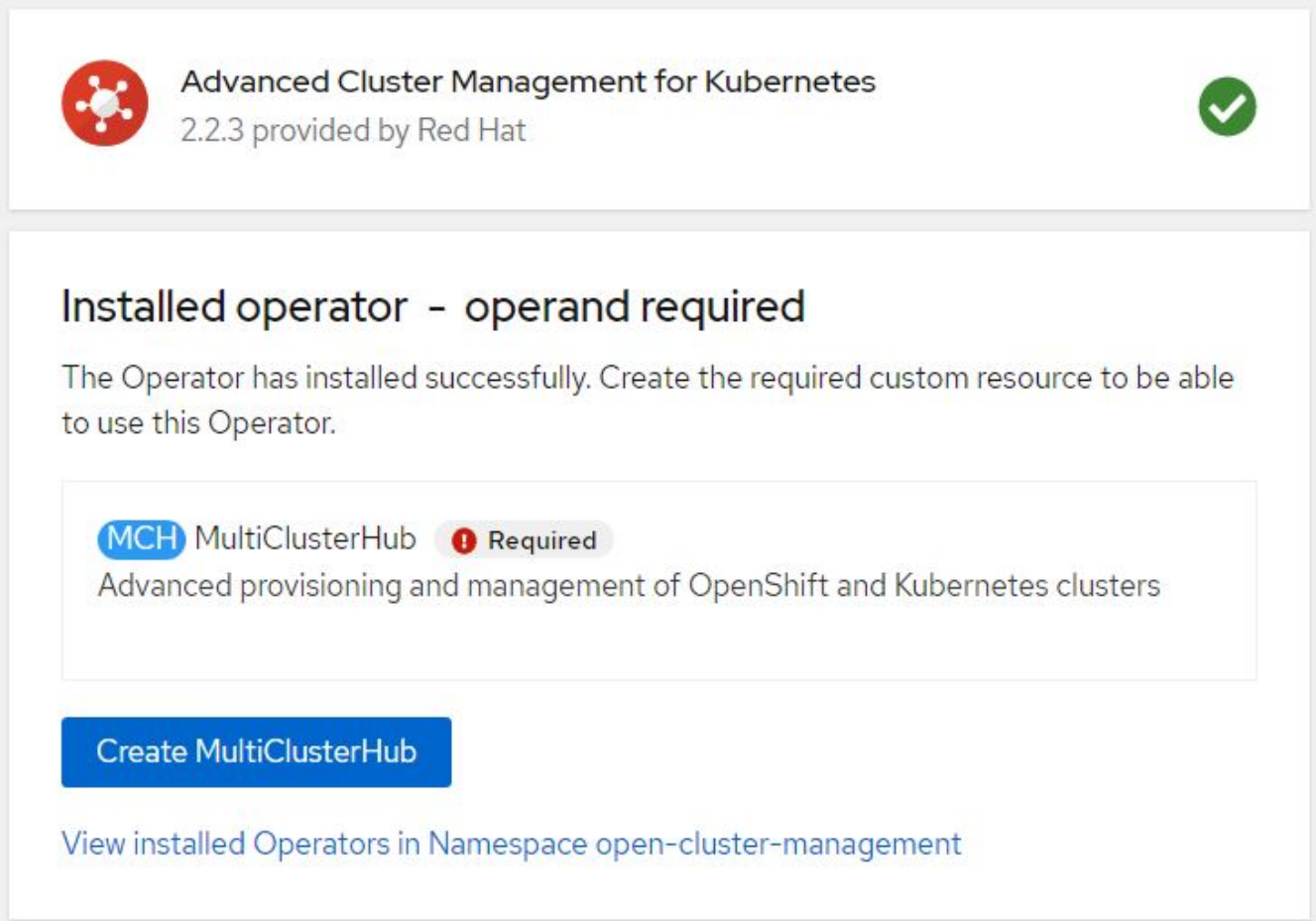
- Automatic
- Manual

Install Cancel

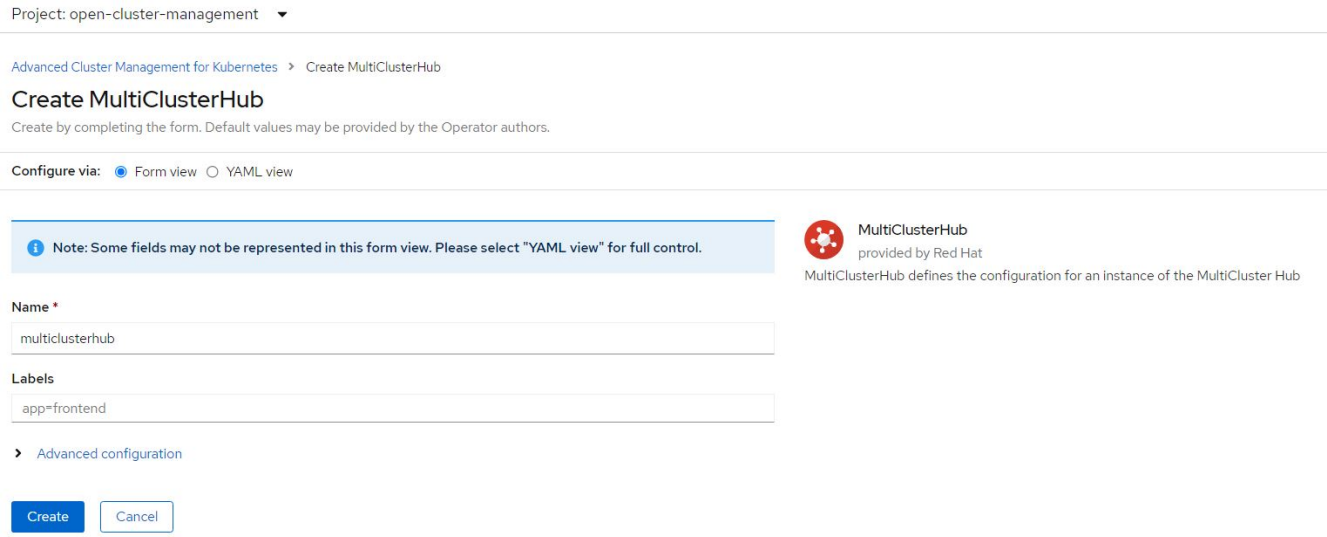
5. Wait for the operator installation to complete.



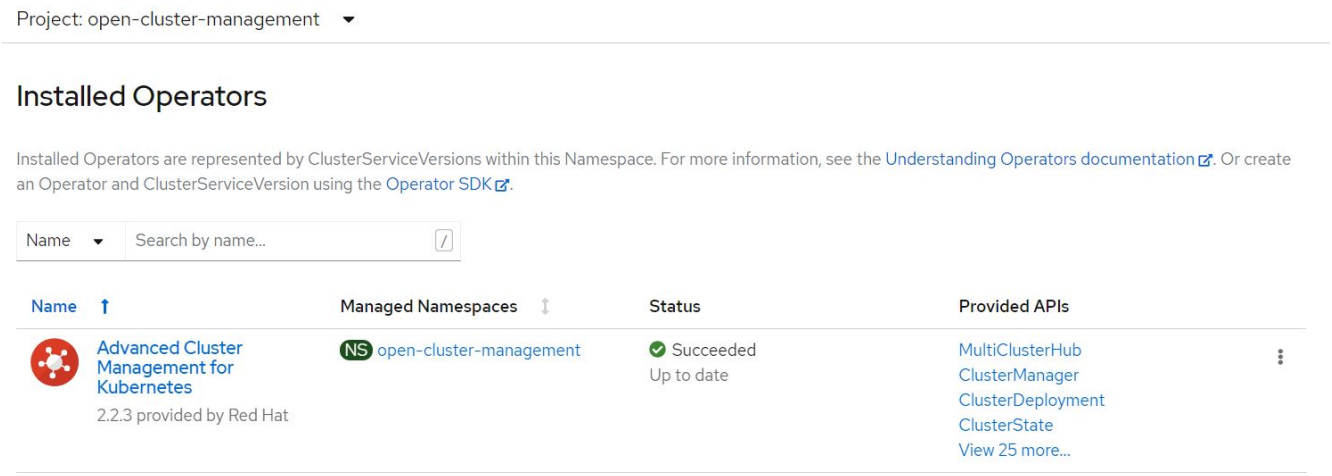
6. After the operator is installed, click `Create MultiClusterHub`.



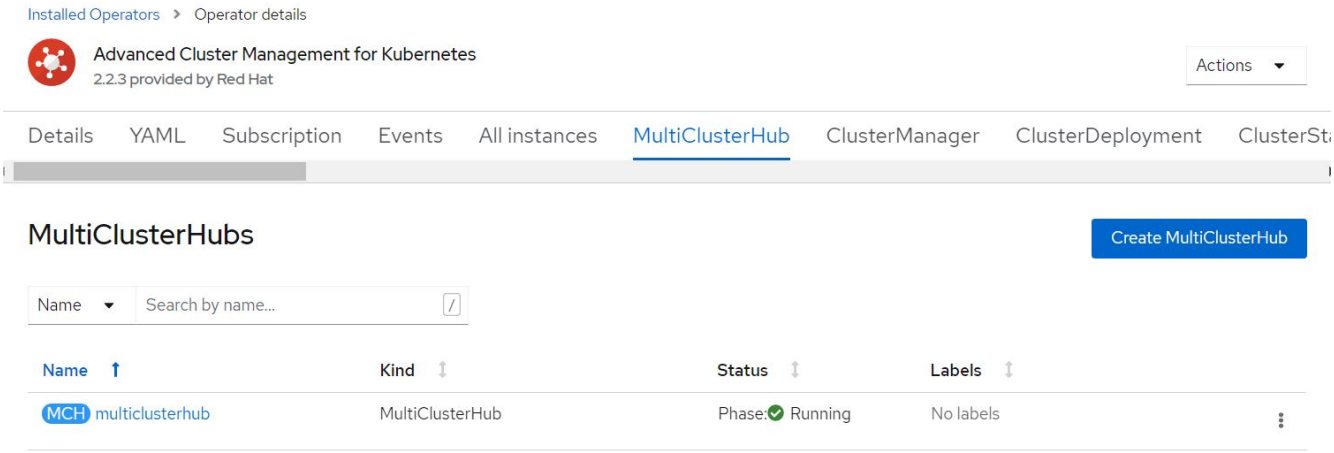
7. On the `Create MultiClusterHub` screen, click `Create` after furnishing the details. This initiates the installation of a multi-cluster hub.



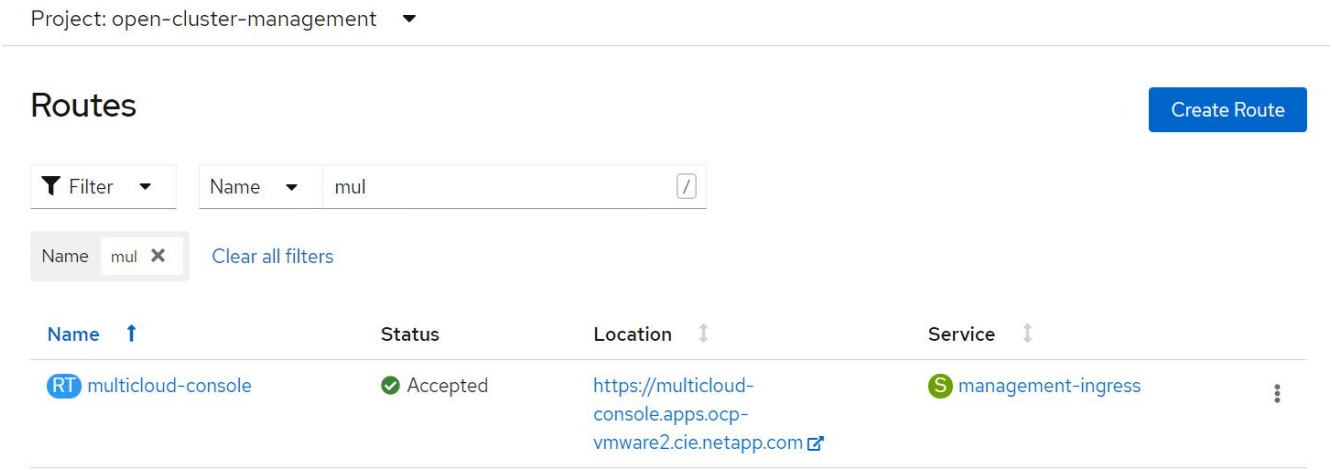
8. After all the pods move to the `Running` state in the open-cluster-management namespace and the operator moves to the `Succeeded` state, Advanced Cluster Management for Kubernetes is installed.



9. It takes some time to complete the hub installation, and, after it is done, the MultiCluster hub moves to `Running` state.



10. It creates a route in the open-cluster-management namespace, connect to the URL in the route to access the Advanced Cluster Management console.



Next: Features - Cluster Lifecycle Management.

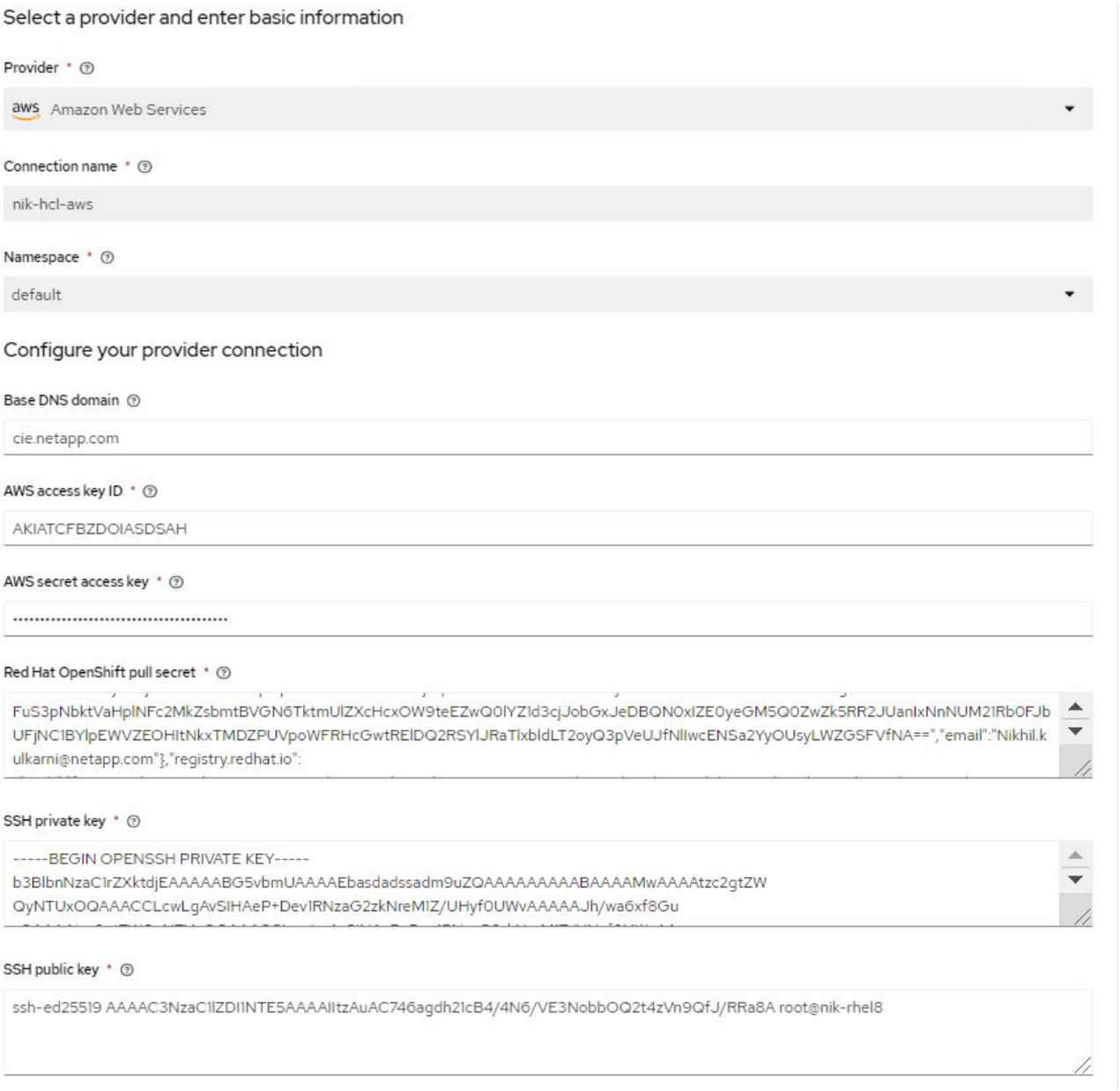
Features

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

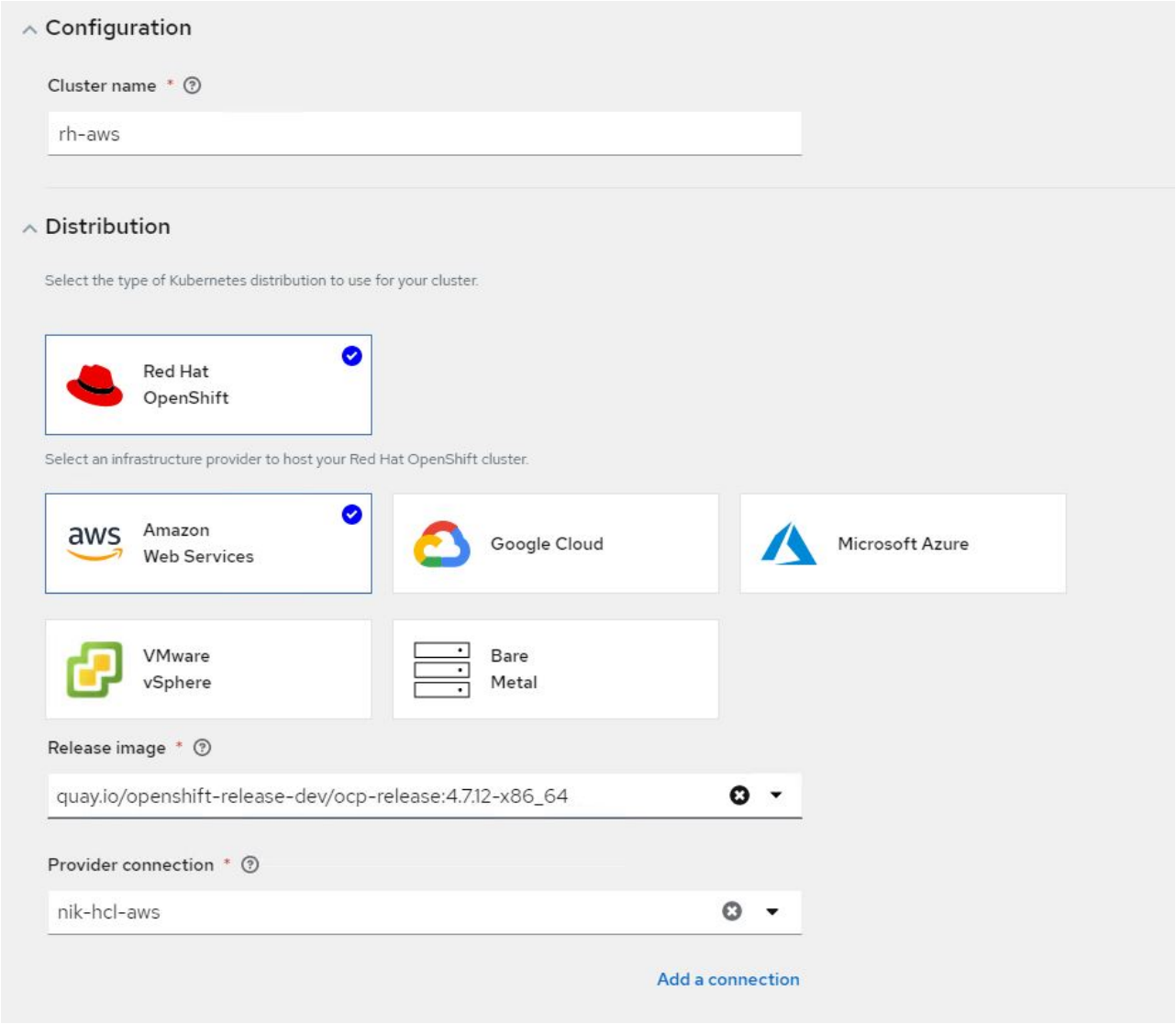
Cluster Lifecycle Management

To manage different OpenShift clusters, you can either create or import them into Advanced Cluster Management.

1. First navigate to `Automate Infrastructures` > `Clusters`.
2. To create a new OpenShift cluster, complete the following steps:
   a. Create a provider connection: Navigate to `Provider Connections` and click on `Add a connection`, provide all the details corresponding to the selected provider type and click on `Add`.



b. To create a new cluster, navigate to `Clusters` and click `Add a cluster` > `Create a cluster`. Provide the details for the cluster and the corresponding provider, and click `Create`.



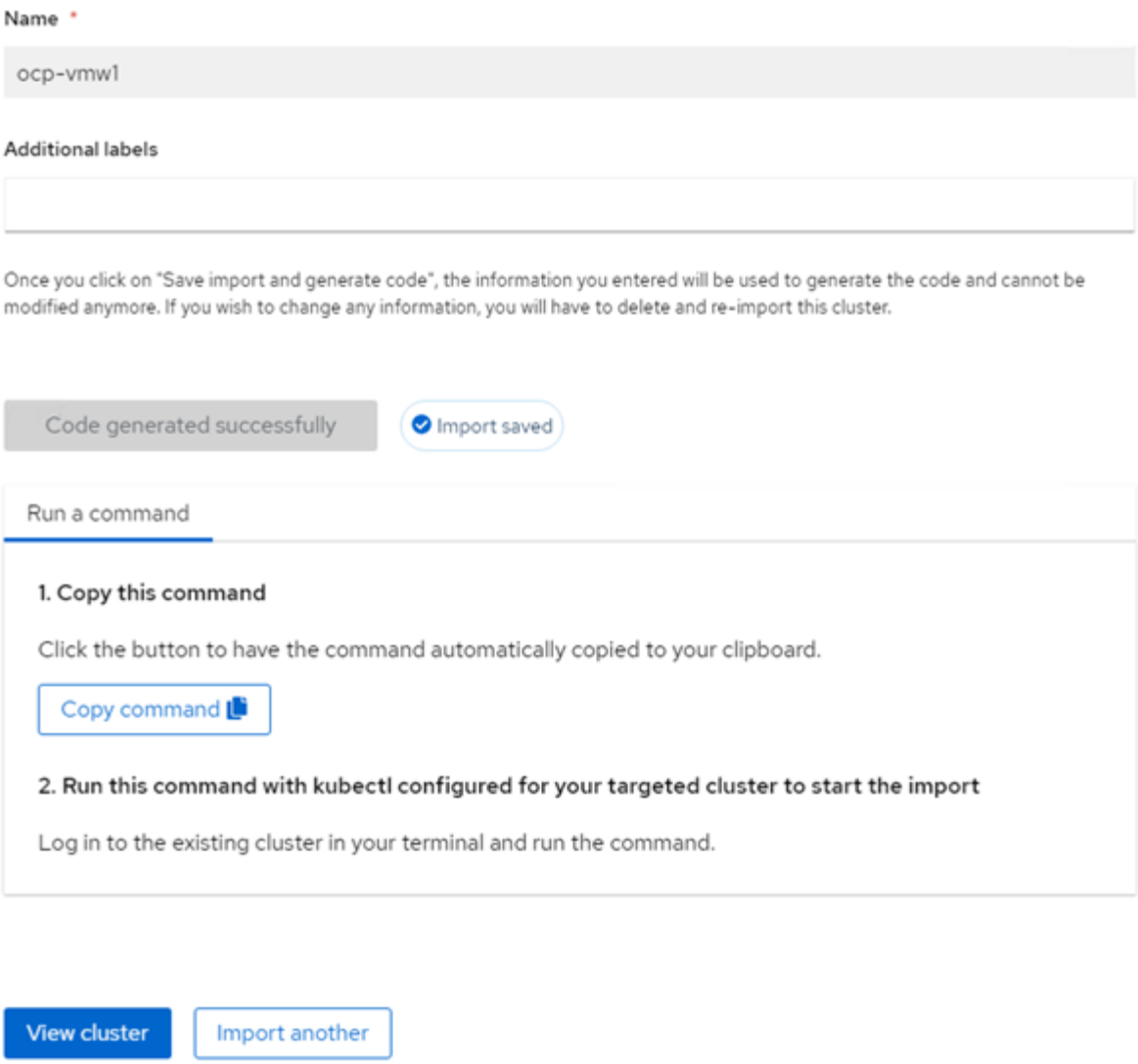
c. After the cluster is created, it appears in the cluster list with the status `Ready`.

3. To import an existing cluster, complete the following steps:

   a. Navigate to `Clusters` and click `Add a cluster` > `Import an existing cluster`.

   b. Enter the name of the cluster and click `Save import and generate code`. A command to add the existing cluster is displayed.

   c. Click `Copy command` and run the command on the cluster to be added to the hub cluster. This initiates the installation of the necessary agents on the cluster, and, after this process is complete, the cluster appears in the cluster list with status `Ready`.



4. After you create and import multiple clusters, you can monitor and manage them from a single console.

Next: Features - Application Lifecycle Management.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Application lifecycle management

To create an application and manage it across a set of clusters,

1. Navigate to `Manage Applications` from the sidebar and click `Create application`. Provide the details of the application you would like to create and click `Save`.

Applications /

Create an application YAML: Off

Cancel Save

Name*

demo-app

Namespace*

default

Repository location for resources

Repository types

Select the type of repository where resources that you want to deploy are located

Git

URL*

https://github.com/open-cluster-management/acm-hive-openshift-releases.git

Branch

main

Path

clusterImageSets/fast/4.7

2. After the application components are installed, the application appears in the list.

Applications

Refresh every 15s

Last update: 7:36:23 PM

Overview Advanced configuration

Create application

Search

Name	Namespace	Clusters	Resource	Time window	Created
demo-app	default	Local	Git		8 days ago

1 - 1 of 1 1 of 1

3. The application can now be monitored and managed from the console.

Next: Features - governance and risk.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Governance and Risk

This feature allows you to define the compliance policies for different clusters and make sure that the clusters adhere to it. You can configure the policies to either inform or remediate any deviations or violations of the rules.

1. Navigate to `Governance and Risk` from the sidebar.
2. To create compliance policies, click `Create Policy`, enter the details of the policy standards, and select the clusters that should adhere to this policy. If you want to automatically remediate the violations of this policy, select the checkbox `Enforce if supported` and click `Create`.

Governance and risk / Policies /

Create policy

YAML: Off

Name *

policy-complianceoperator

Namespace *

default

Specifications *

1 ✕ ComplianceOperator

Cluster selector

1 ✕ local-cluster: "true"

Standards

1 ✕ NIST-CSF

Categories

1 ✕ PR.IP Information Protection Processes and Procedures

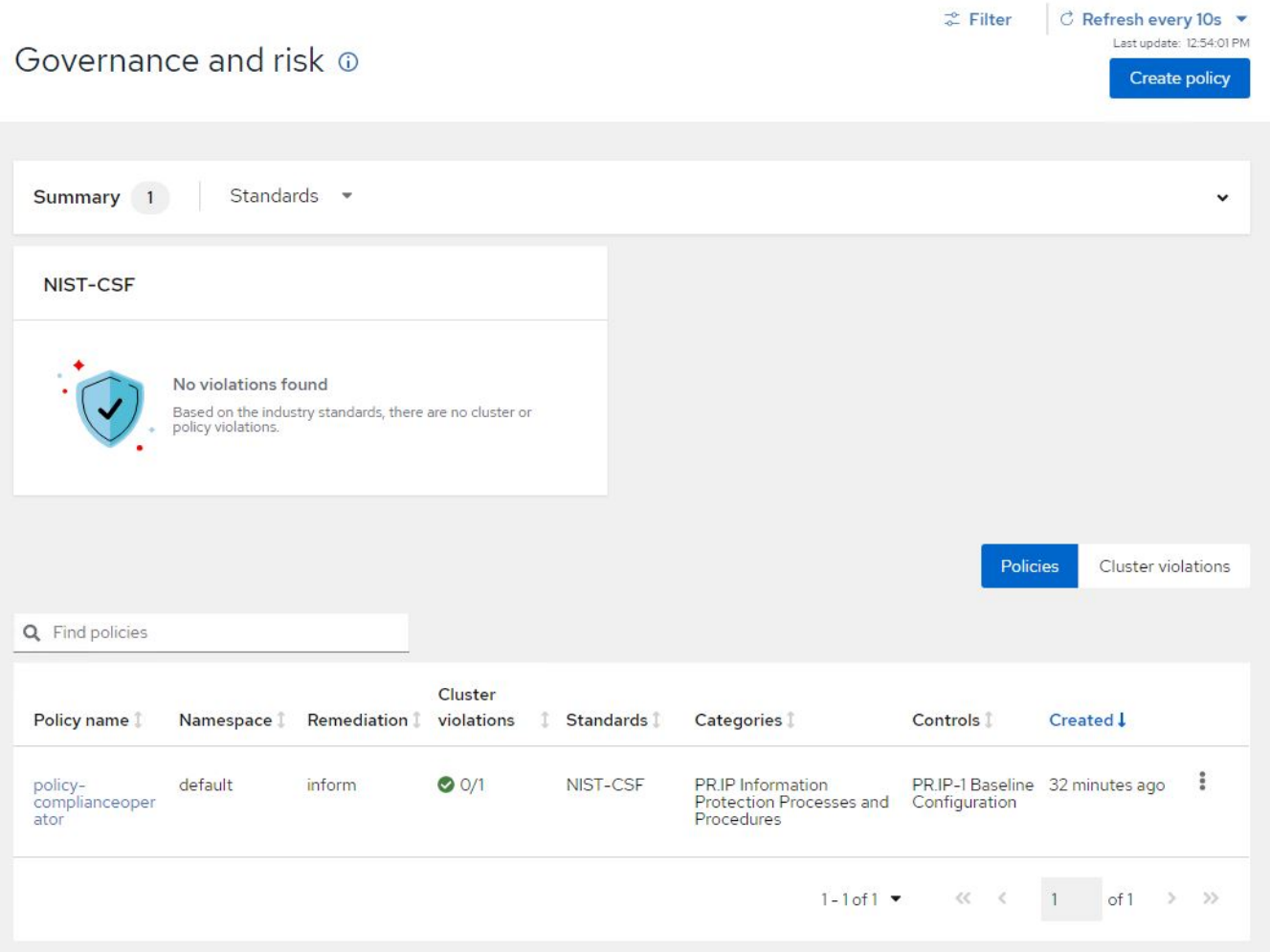
Controls

1 ✕ PR.IP-1 Baseline Configuration

☐ **Enforce if supported**

☐ **Disable policy**

3. After all the required policies are configured, any policy or cluster violations can be monitored and remediated from Advanced Cluster Management.



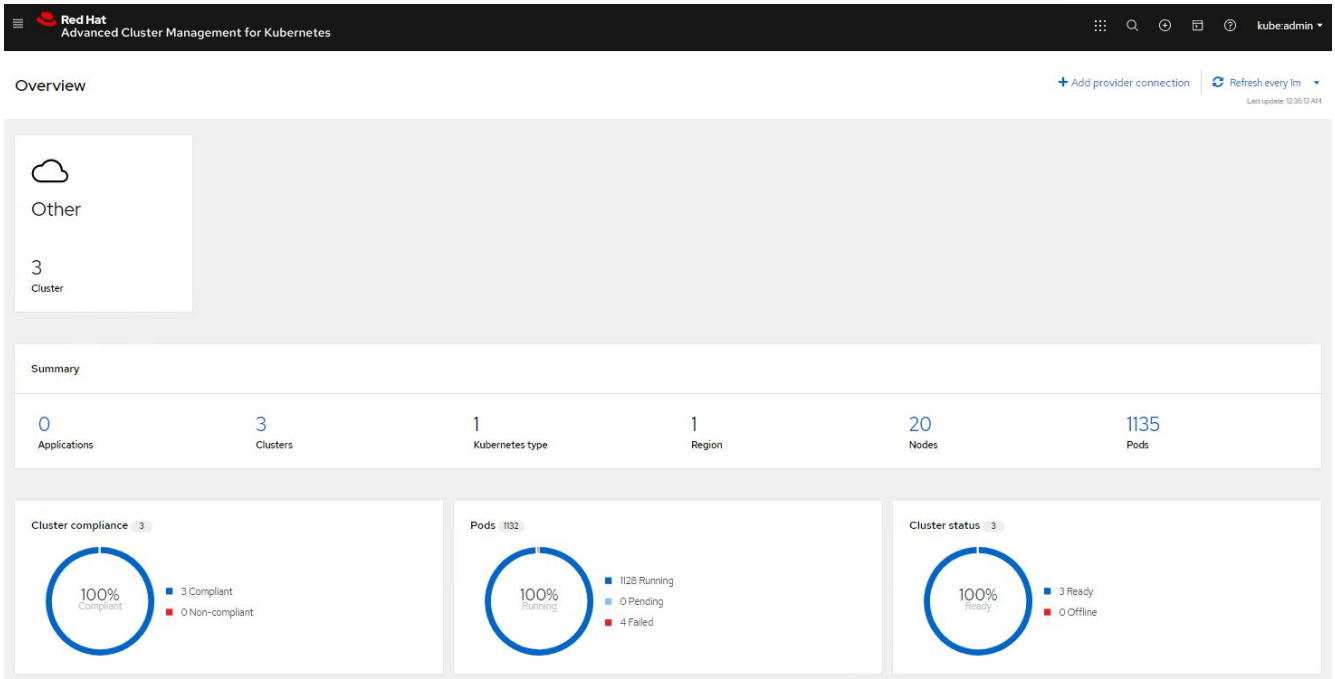
Next: Features - Observability.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

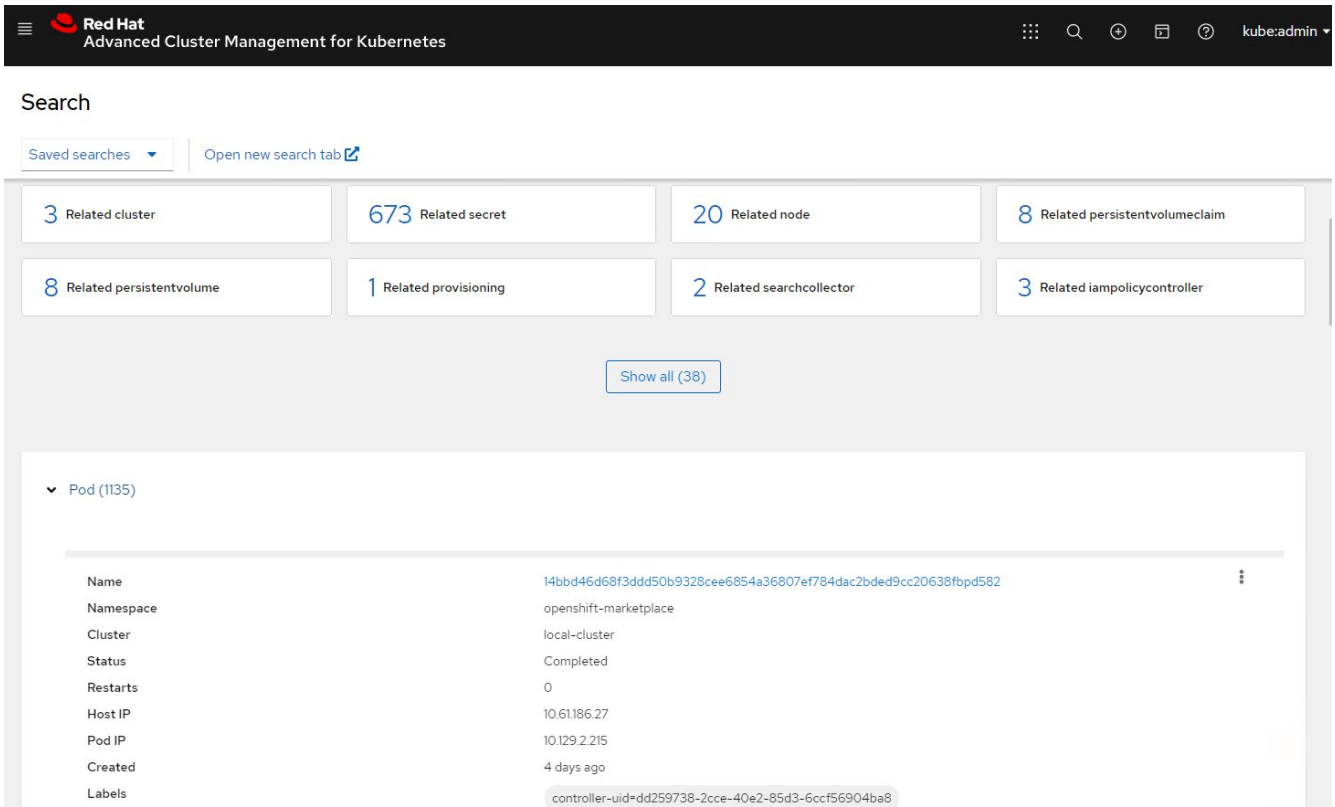
Observability

Advanced Cluster Management for Kubernetes provides a way to monitor the nodes, pods, and applications and workloads across all the clusters.

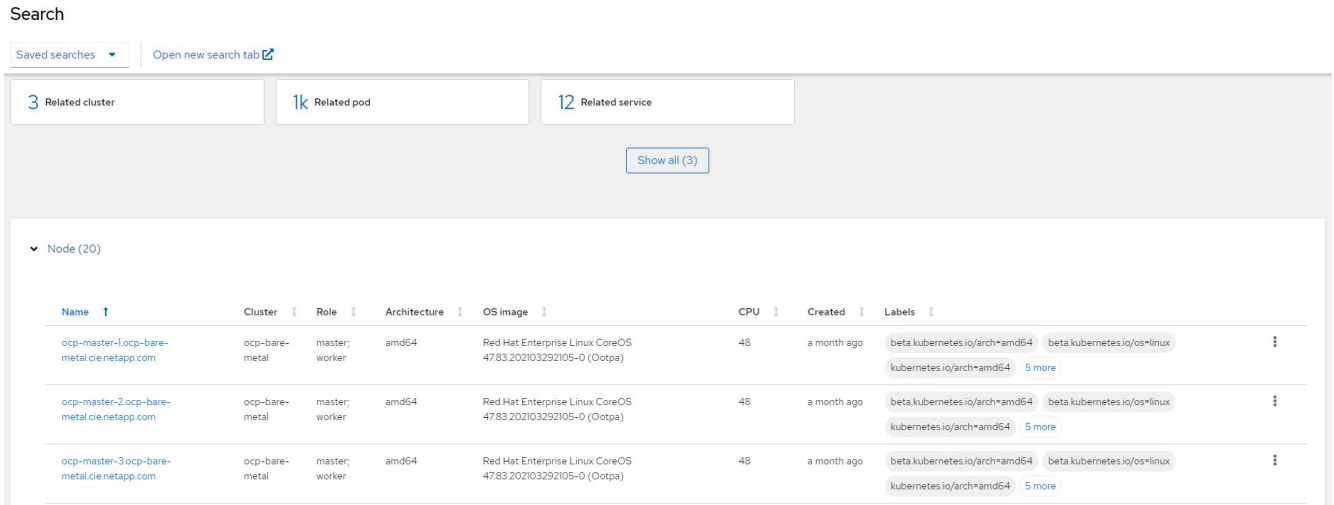
1. Navigate to `Observe Environments` > `Overview`.



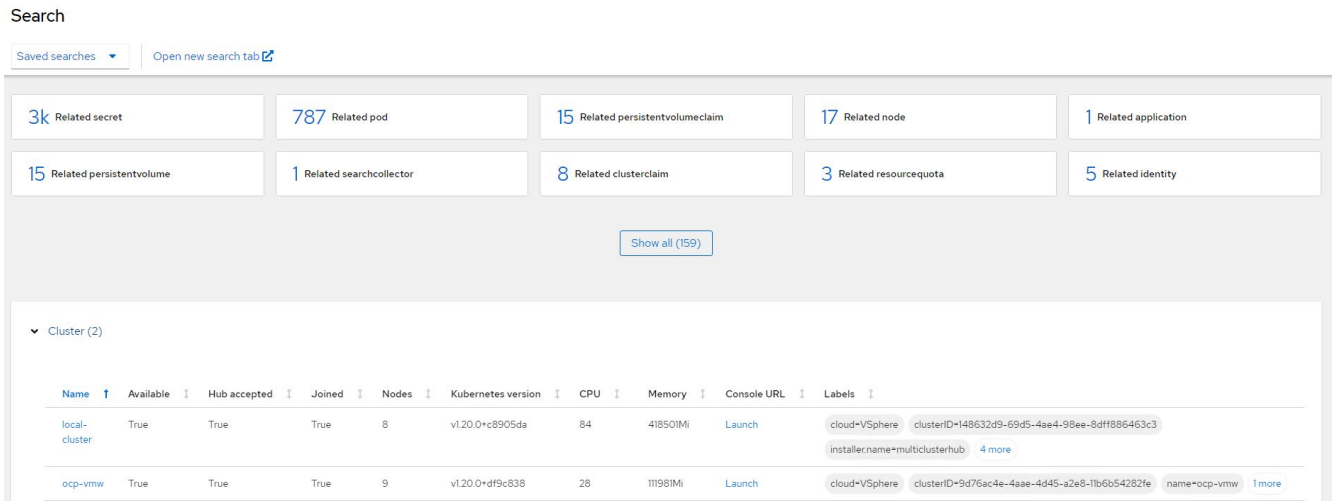
2. All pods and workloads across all clusters are monitored and sorted based on a variety of filters. Click `Pods` to view the corresponding data.



3. All nodes across the clusters are monitored and analyzed based on a variety of data points. Click `Nodes` to get more insight into the corresponding details.



4. All clusters are monitored and organized based on different cluster resources and parameters. Click `Clusters` to view cluster details.



Next: Features - Create Resources.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Create resources on multiple clusters

Advanced Cluster Management for Kubernetes allows users to create resources on one or more managed clusters simultaneously from the console. As an example, if you have OpenShift clusters at different sites backed with different NetApp ONTAP clusters, and want to provision PVC's at both sites, you can click the `+` sign on the top bar. Then select the clusters on which you want to create the PVC, paste the resource YAML, and click `Create`.

Create resource

Cancel Create

Clusters | Select the clusters where the resource(s) will be deployed.

2 × local-cluster, ocp-vmw

Resource configuration | Enter the configuration manifest for the resource(s).

YAML

```yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: demo-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: ocp-trident
```

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